

ENCYCLOPEDIA
OF ASTRONOMY
AND ASTROPHYSICS

NATURE PUBLISHING

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Olympus Mons

The largest volcano on Mars, and the largest in the solar system, centered at 18.4°N , 133.1°W . It coincides with the albedo feature visible from Earth known as Nix Olympica, the 'Snows of Olympus', after the mountain in Greece which in legend was the home of the gods. Olympus Mons rises to a height of 27 km above Mars's mean surface level, and in its greatest horizontal dimension measures 624 km. The central caldera has a diameter of about 90 km; from there, undulating flanks slope away at an average gradient of 4° toward the peripheral cliff, whose walls tower up to 6 km above the surrounding plateau. This huge volcanic edifice contains a hundred times the mass of the Earth's largest volcano, Hawaii's Mauna Loa, which is 120 km long and rises to just over 9 km above the Pacific Ocean floor. On the Earth, plate tectonics operates to carry a volcano away from the underlying source of magma. With no such movement on Mars, volcanoes could continue to grow for as long as magma was available.

See also: Mars: surface.